



Department of Computer Science and Business Systems

Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch : IV Semester B.Tech CSBS
Course Code & Title : CS3492 Database Management Systems
Name of the Faculty member : Dr.S.Erana Veerappa Dinesh, ASP / CSBS

Innovative Practice Description

- **Unit / Topic:** Unit III / Deadlock Handling
- **Course Outcome:** CO3
- **Activity Chosen:** Crossword puzzle
- **Justification:**
 - ✓ Deadlock handling is a crucial topic in advanced database systems and operating systems. It refers to the problem that arises when a group of transactions are each waiting for a resource held by the other, creating a cycle of dependency where no transaction can proceed. Understanding how to prevent, detect, and recover from deadlocks is vital to ensure the consistency and availability of database systems.
 - ✓ A crossword puzzle is a word game that involves filling a grid with words based on clues provided. It engages the learner in critical thinking and helps reinforce terminology and concepts in a fun and interactive way.
 - ✓ Therefore, a crossword puzzle activity is chosen to enhance conceptual clarity of deadlock handling techniques, such as wait-die, wound-wait, deadlock detection algorithms, and recovery strategies, by identifying the correct terms and understanding their context through clue-based learning.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 - Faculty gave a brief overview of deadlock concepts and explained its working procedure & types.
 - Students were given few minutes to individually think about their understanding of deadlock detection and prevention concepts.
 - A crossword puzzle is a fun word game that encourages students to remember the idea and try to figure out the solution on their own. Instead of administering standard MCQ tests, it sparks pupils' curiosity.
 - The activity was conducted in a classroom for duration of 20 minutes. The students were given 10 questions as a crossword puzzle in the topic of email integration. They have to think and write the answers in the given puzzle.



- The activity was conducted during the last 20 minutes of the class after that the answers were corrected. Sample answers were shown in figure 1. The answers for the wrong questions were discussed at the end of the class to make them to know the answer of all the questions.
- CO-PO/ PSO mapping:**

CO	PO1	PO2	PO3	PO10	PO12	PSO1
CO2	3	2	1	2	2	2

(1– Low 2 –Moderate 3 – High)

- PO/ PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PSO1
	3	3	1	2	2
Justification for correlation	The students apply the knowledge of deadlock	The students analyze the given questions and identify solutions	The students find the solutions for the given question.	The students demonstrate problem-solving skills as an individual.	The students apply the concepts of deadlock for solving the novel problems

- Images / Screenshot of the practice:**

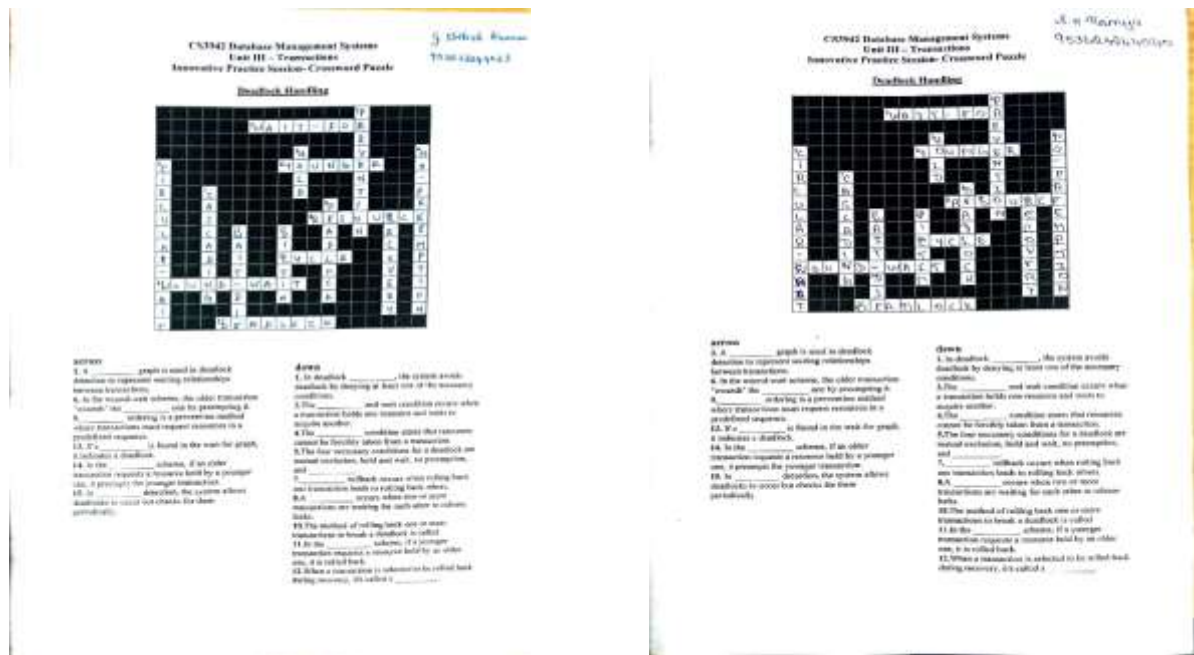


Fig1: Sample Answers by Nithish Kannan G and Ramya S K of II year CSBS



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- The crossword puzzle activity was very interesting and able to identify the appropriate name of the dead handling methods.
- It will be helpful while preparing for examination.

- ❖ ***Benefit of the practice:***

- Student's understanding on deadlock handling is improved.
- Vocabulary of the terms related to the topic is improved.
- This activity helps to test the level of understanding of the students

- ❖ ***Challenges faced in implementation:***

- Few students found it difficult to complete the puzzle
- Students just might not have the necessary knowledge to complete crossword puzzles.

References:

- https://www.ritrjpm.ac.in/images/CSBS/2023-2024/CWP_GE3151_PSPP_BY_23_24.pdf
- https://www.ritrjpm.ac.in/images/CSBS/2023-2024/CW_GE3151_PSPP_KU_23_24.pdf
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- <https://www.ritrjpm.ac.in/images/computer-science/Crosswords-MCAP-GM.pdf>
- <http://www.classtools.net/crossword/>
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